Signal Processing • BA4

1 Basic things

Unit impulse | Delta

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}, \qquad \delta_m[n] = \delta[n-m]$$

Unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Exponential decay

$x[n] = a^n u[n], \quad 0 < a < 1$

Sinusoidal oscillation

$x[n] = A\cos(\omega n + \phi), \quad A \in \mathbb{R},\ \omega \in \mathbb{R},\ \phi \in \mathbb{R}$

Complex exponential

$x[n] = A\, e^{j(\omega n + \phi)}, \quad A \in \mathbb{R},\ \omega \in \mathbb{R},\ \phi \in \mathbb{R}$

Energy and Power

- $E_x = \|x\|^2 \quad$ (sum or integral of $|x|^2$)
- $P_x = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$

Operator combinations

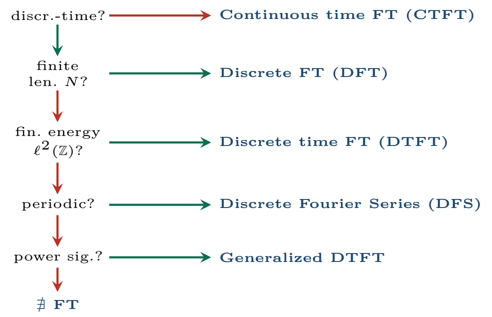
signal	sample at time n
$\mathcal{R}\mathbf{x}$	$x[-n]$
$\mathcal{S}^d\mathbf{x}$	$x[n+d]$
$\mathcal{R}\mathcal{S}^d\mathbf{x}$, $\mathcal{S}^{-d}\mathcal{R}\mathbf{x}$	$x[-n+d]$
$\mathcal{R}\mathcal{S}^{-d}\mathbf{x}$, $\mathcal{S}^d\mathcal{R}\mathbf{x}$	$x[-n-d]$

Sum of a sequence

$\sum_{k=m}^{n} k = \frac{(m+n)(n-m+1)}{2}$

$\sum_{k=m}^{n} z^k = \begin{cases} n-m+1 & z = 1 \\ \frac{z^m - z^{n+1}}{1-z} & z \neq 1 \end{cases}$

2 Signal analysis



3 DFT: $X[k]$

Fourier basis vectors

$w_k[n] = e^{j\frac{2\pi}{N}kn}, \quad n, k = 0, \ldots, N-1.$
Note: $\mathbf{w}_k = \mathbf{w}_{N-k}^*$.

Orthogonality (not orthonormality)

$\langle \mathbf{w}_h, \mathbf{w}_k \rangle = N\,\delta[h-k]$, where

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x^*[n]\, y[n].$$

DFT matrix form

$\mathbf{X} = \mathbf{W}\mathbf{x}, \quad \mathbf{x} = \frac{1}{N}\mathbf{W}^H\mathbf{X}$, where $W_N = e^{-j\frac{2\pi}{N}}$:

$$\mathbf{W} = \begin{bmatrix} W_N^0 & W_N^0 & \cdots & W_N^0 \\ W_N^0 & W_N^1 & \cdots & W_N^{N-1} \\ \vdots & & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & \cdots & W_N^{(N-1)^2} \end{bmatrix}$$

$\mathbf{W}^H\mathbf{W} = N\mathbf{I}$.
$W_N^n = W_N^{n \bmod N}$ (aliasing property).

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

4 Properties

A signal is **Hermitian-symmetric** if $x[n] = x^*[N-n]$.
DFS $\equiv$ DFT: same formulas, different interpretation (explicit periodicity).

5 DTFT: $X(e^{j\omega})$

Symmetry summary

Time domain $x[n]$	**Freq. domain** $X(\omega)$
Real	$X(-\omega) = X^*(\omega)$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

Generalized DTFT (Power Signals)

TODO [Explain the difference]

6 FFT

TODO [Explain the difference]

7 Spectrograms (STFT)

TODO [Explain the difference]

8 Filtering I - Digital Filters

LTI: Linearity & time-invariance

Let $y[n] = \mathcal{H}\{x[n]\}$.
Linearity ($\forall\, \alpha, \beta \in \mathbb{C}, \forall\, x_1, x_2$):

$$\mathcal{H}\{\alpha\, x_1[n] + \beta\, x_2[n]\} \stackrel{?}{=} \alpha\, \mathcal{H}\{x_1[n]\} + \beta\, \mathcal{H}\{x_2[n]\}$$

Time-Invariance ($\forall\, n_0 \in \mathbb{Z},\ \forall\, x$):

$$\underbrace{\mathcal{H}\{x[n-n_0]\}}_{x[\cdot]\leftarrow x[\cdot-n_0]} \stackrel{?}{=} \underbrace{y[n-n_0]}_{n\leftarrow n-n_0}$$

Impulse response

$\mathbf{h} = \mathcal{H}\boldsymbol{\delta}$. An LTI system is *fully* described by $\mathbf{h}$.

Convolution $\mathbf{y} = \mathbf{x} * \mathbf{h}$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Convolution properties

Commutativity: $\mathbf{x} * \mathbf{h} = \mathbf{h} * \mathbf{x}$
Linearity: $\mathbf{x} * (\alpha\mathbf{y} + \beta\mathbf{w}) = \alpha(\mathbf{x} * \mathbf{y}) + \beta(\mathbf{x} * \mathbf{w})$
Time-shift: $\mathbf{x} * (\mathcal{S}^k\mathbf{y}) = \mathcal{S}^k(\mathbf{x} * \mathbf{y})$
Associativity: $(\mathbf{x} * \mathbf{h}) * \mathbf{w} = \mathbf{x} * (\mathbf{h} * \mathbf{w})$
Cascade: $\mathbf{h} = \mathbf{h}_1 * \mathbf{h}_2$; order is irrelevant.

FIR vs IIR (Finite/Infinite Impulse Response)

- **FIR**: $\mathbf{h}$ finite support. No feedback; *always* BIBO stable.
- **IIR**: $\mathbf{h}$ infinite support.

Definition of causal

Only responds to present or past inputs.

BIBO Stability

LTI is BIBO stable $\iff$ $\mathbf{h}$ is absolutely summable

Frequency Response

We extract from it the magnitude/phase response.

$$H(\omega) = \text{DTFT}\{\boldsymbol{h}[n]\}$$

Filter types by magnitude response

- **Lowpass**: passes low freqs ($\omega \approx 0$)
- **Highpass**: passes high freqs
- **Bandpass**: passes freqs near $\pm\omega_c$
- **Allpass**: $|H(\omega)| = \text{const}$ — modifies phase only

Stop Band: 0 - Transition Band - Pass Band: 1.

Example of filters

Moving Average (MA)

$$y[n] = \frac{1}{M}\sum_{k=0}^{M-1} x[n-k]$$

$h[n] = \frac{1}{M}$ for $0 \le n < M$, else 0. Causal FIR $\Rightarrow$ always BIBO stable. Big cost.

Leaky Integrator (LI)

$$y[n] = \lambda\, y[n-1] + (1-\lambda)\, x[n], \quad |\lambda| < 1$$

$$h[n] = (1-\lambda)\lambda^n\, u[n] \quad \text{(causal, exponential decay)}$$

Causal IIR. Constant cost. Approximates MA of length M with $\lambda = \frac{M-1}{M}$.

9 Filtering II - Filter Design

Transfer function $H(z)$

$$H(z) = \frac{B(z)}{A(z)} = b_0\, \frac{\prod_{k=0}^{M-1}(1 - z_k z^{-1})}{\prod_{k=0}^{N-1}(1 - p_k z^{-1})}$$

$B(z)$: feedforward (FIR part).
$A(z)$: feedback (IIR part). FIR: $A(z) = 1$.
Zeros z_k: roots of $B(z)$, where $|H| = 0$.
Poles p_k: roots of $A(z)$, where $|H| \to \infty$.

ROC & causality

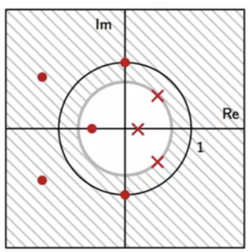
$$z \in \text{ROC}\{X(z)\} \iff \sum_{n=-\infty}^{\infty} \left|x[n]z^{-n}\right| < \infty$$

Causal IIR: ROC $= \{|z| > R\}$: $R = \max_k |p_k|$.
FIR: ROC $= \mathbb{C}$.

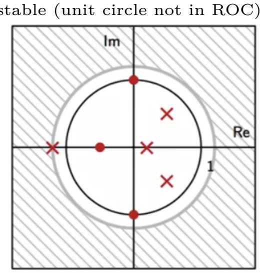
BIBO stability

Stable $\iff$ unit circle $|z| = 1$ lies inside the ROC $\iff$ all poles strictly inside the unit circle ($|p_k| < 1$).

Stable (unit circle in ROC)



Unstable (unit circle not in ROC)



10 Filtering III - Stochastic and Adaptive SP

Reminder - Proba

CDF: $F_X(\alpha) = P(X \le \alpha)$
PDF: $f_X(\alpha) = \frac{d}{d\alpha}F_X(\alpha)$
Moments: $E[g(x)] = \int_{-\infty}^{\infty} g(x)\, f_x(x)\, dx$
Variance: $\sigma_x^2 = E[(x-\mu_x)^2]$

Cross-correlation & Covariance

Cross-correlation: $R_{xy} = E[xy]$
Covariance: $C_{xy} = E[(x-m_x)(y-m_y)]$
Uncorrelated: $E[xy] = E[x]E[y] = m_x m_y$
Independent: $f_{xy}(x,y) = f_x(x) f_y(y)$

Cross-Correlation

$$r_{xy}[k] = \langle \mathbf{x}, \mathcal{S}^k\mathbf{y}\rangle = \sum_{n=-\infty}^{\infty} x^*[n]\, y[n+k]$$

White Noise Process

Zero-mean: $E[x[n]] = 0$
Uncorr.: $E[x[n]x[m]] = E[x[n]]E[x[m]]$ for $m \neq n$
Autocorrelation: $r_x[n] = \sigma_x^2\, \delta[n]$
PSD: $P_w(\omega) = \sigma_w^2$

Stationarity

Statistical properties are time-invariant. In practice, we use **WSS**, which only requires the first two moments to be time-invariant.

Wide-Sense (WSS) stationarity

For WSS random processes we only care about the first two moments:

- $E[x[n]] = m_x$
- $E[x[n]\, x[m]] = r[n-m]$

Autocorrelation (not given)

$$x[n] = a^n u[n] \;\Rightarrow\; r_x[k] = \frac{a^{|k|}}{1-a^2}$$

$$\text{DTFT}\{r_x[k]\} = \frac{1}{1 + a^2 - 2a\cos\omega}$$

Power Spectral Density (PSD)

Estimator (DFT): $P[k] = E\left[|X_N[k]|^2/N\right]$
Exact definition (Wiener-Khinchin):

$$P_x(\omega) = |X(\omega)|^2 = \text{DTFT}\{r_x[k]\}$$

Variance: $\sigma_x^2 = \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} P_x(\omega)\, d\omega = r_x[0]$

Convolution theorem (stochastic)

If $y[n] = h[n] * x[n]$, then: $P_y(\omega) = |H(\omega)|^2 P_x(\omega)$

11 A/D I - Interpolation & Sampling

The Continuous-Time World

CT vs. DT paradigm

D: $x_d[n] \in \ell_2(\mathbb{Z})$, $n \in \mathbb{Z}$, $\omega \in [-\pi, \pi]$, DTFT
A: $x_c(t) \in L_2(\mathbb{R})$, $t \in \mathbb{R}$ [s], $f \in \mathbb{R}$ [Hz], CTFT

Use cases

- x(t) → **[processing]** → y(t)
- x[n] → **[processing for synthesis]** → y(t)
- x(t) → **[processing for analysis]** → y[n]

Real-world frequency

f = #rep/s = Hz ; $\Omega = 2\pi f$

CTFT (more info in cs)

- NOT PERIODIC
- NO LIMIT FREQUENCY

BandLimited (BL) functions

A CT signal is F_S-BL ⇔ X(f) for $|f| > F_S/2$

Support of a function

The interval where the function is non-trivially zero

Sampling theorem (Shannon–Nyquist)

Any signal $x(t)$ whose highest frequency component is F_N Hz can be sampled with **no loss of information** using a sampling frequency:

$$F_s \geq 2F_N \quad \left(\text{i.e. } T_s \leq \frac{1}{2F_N}\right)$$

Interpolation x[n] → x(t)

Interpolation requirement

- decide on T_s (sampling period)
- make sure x(nT_s) = x[n]
- make sure x(t) is smooth

Interpolations pros and cons

Polynomial: + C^∞ (max. smooth) − kernels $L_n^{(N)}$ depend on N
Local: + kernel indep. of N − lacks smoothness

Remarkable result

$$\lim_{N\to\infty} L_n^{(N)}(t) = \text{sinc}(t-n)$$

Sinc interpolation

$$x(t) = \sum_{n\in\mathbb{Z}} x[n]\,\text{sinc}(t-n)$$

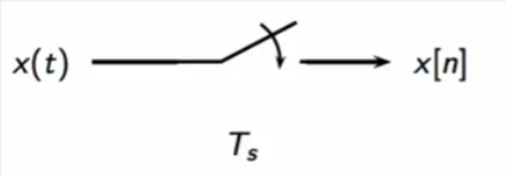
$\text{sinc}(m-n) = \delta[m-n], \forall m, n \in \mathbb{Z}$
CTFT pair ($T_s = 1/F_s$):

$$\varphi(t) = \text{sinc}\left(\frac{t}{T_s}\right) \longleftrightarrow \Phi(f) = \frac{1}{F_s}\,\text{rect}\left(\frac{f}{F_s}\right)$$

12 A/D II - Aliasing

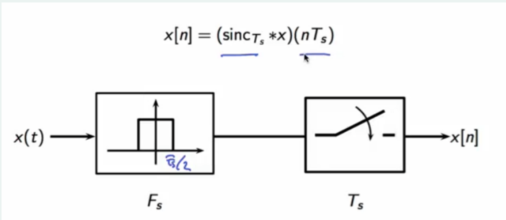
Raw Sampling

$$x[n] = x(nT_s)$$



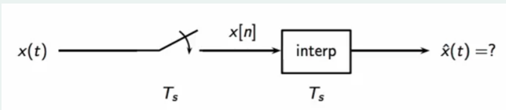
Sinc Sampling

$$x[n] = \left\langle \text{sinc}\left(\frac{t-nT_s}{T_s}\right), x(t)\right\rangle$$



Aliasing

Sinc Sampling



Map the frequency into $[-F_s/2\ ;\ F_s/2]$ to make the signal F_s-BL.

13 A/D III - Multirate

Downsampling by N "DTFT pov"

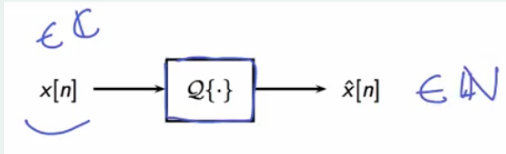
Copier à tout 2 pi/N. Multiplier les indices par N.

Upsampling by N "DTFT pov"

Montrer la périodiciter. Diviser les indices par N.

14 A/D IV - A/D and D/A conversion

Quantization schemes



Quantization MSE

$$\sigma_e^2 = E\left[(x - Q(x))^2\right]$$

Error analysis for uniform quantization

Error energy: $\sigma_e^2 = \Delta^2/12$, $\quad \Delta = (B-A)/2^R$
Signal energy: $\sigma_x^2 = (B-A)^2/12$
SNR: $\text{SNR} = \sigma_x^2/\sigma_e^2 = 2^{2R}$
$\textbf{SNR}_{\textbf{dB}}$: $10\log_{10} 2^{2R} \approx 6R$ dB
A linear filter cannot improve the SNR due to quantization if the quantizer matches the input.